# Chen Hajaj

List of Publications

*Updated October 2026*

## REFEREED JOURNAL PUBLICATIONS

1. Udi Aharon, Ran Dubin, Amit Dvir, and **Chen Hajaj**. HTTP2DoS: Detecting low-and-slow HTTP/2 DoS attacks with complex transfer-entropy graph-temporal learning. *Computers & Security*. 2026
2. Yehonatan Zion, Amit Dvir, **Chen Hajaj**, Ran Dubin, and Neriya Filber. Ultra Fast Encrypted Network Traffic Classification Based on Classification by Retrieval Powered by Out-of-Distribution. *Computer Networks*, 289:112739. 2026
3. Matan Klein, Revital Marbel, **Chen Hajaj**, Ran Dubin, and Amit Dvir. GraphMux: A graph-based framework for encrypted traffic classification. *Computer Networks*. 2026
4. **Chen Hajaj**, Shani Alkoby, Shai Ashkenazi, Avner Herman Cohen, Yael Reichenberg, Shiri Kushnir, Vered Shkalim Zemer. Machine Learning Tools for Predicting Pediatric Urinary Tract Infections Caused by ESBL-producing Bacteria. *The Pediatric Infectious Disease Journal*. 2026
5. Zohar Simhon, Matan Weiss, Revital Marbel, **Chen Hajaj**, Amit Dvir, Ran Dubin. Real-Time Network Security: Integrating ANN and Dynamic Graph-Based Clustering. *Computer Networks*. 2026
6. Ron Hirshprung, Ori Lesman, **Chen Hajaj**. The Sound of Emotions, An Artificial Intelligence Approach to Predicting Emotions from Musical Selections. *Multimedia Systems*. 2026
7. Anat Goldstein, **Chen Hajaj**, Pierre Singer, Orit Raphaeli. Survival Forest Models for ICU Mortality Prediction based on Nutrition and Clinical Factors. *IEEE Access*. 2026
8. Zafrir Avrahami, Moti Zwilling, **Chen Hajaj**. Leveraging OSINT for Advanced Proactive Cybersecurity: Strategies and Solutions. *IEEE Access*. 2025
9. Ro'i Lang, Or Haim Anidjar, Sahar Slonimsky, **Chen Hajaj**, Oz Golan, Carmel Matias, Alex Diskin, Evgeny Strokin, Mor Mega. Measuring and Analyzing Defects of Additive Manufactured Ti-6Al-4V Specimens Through Image Segmentation. *Fatigue & Fracture of Engineering Materials & Structures*. 2025
10. Udi Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. A classification-by-retrieval framework for few-shot anomaly detection to detect API injection. *Computers & Security*. 2025
11. Or Haim Anidjar, Revital Marbel, Ran Dubin, Amit Dvir, **Chen Hajaj**. Extending Limited Datasets with GAN-Like Self-Supervision for SMS Spam Detection. *Computers & Security*. 2024
12. Assaf Frajman, Shimon Izhakian, Ori Mekiten, Ori Hadar, Ariel Lichtenstadt, **Chen Hajaj**, Shon Schori, Moshe Heching, Dror Rosengarten and Mordechai R. Kramer. Phenotypical Characteristics of Nontuberculous Mycobacterial Infection in Patients with Bronchiectasis. *Respiratory Research*. 2024
13. **Chen Hajaj**, Porat Aharon, Ran Dubin, and Amit Dvir. The Art of Time-Bending: Data Augmentation and Early Prediction for Efficient Traffic Classification. *Expert Systems with Applications*. 2024
14. Anat Goldstein and **Chen Hajaj**. Measuring flight-destination similarity: A multidimensional approach. *Expert Systems with Applications*. 2024
15. Ofek Bader, Adi Lichy, Amit Dvir, Ran Dubin, and **Chen Hajaj**. OSF-EIMTC: An Open-Source Framework for Standardized Encrypted Internet Traffic Classification. *Computer Communications*. 2024
16. Orit Raphaeli, Liran Statlender, **Chen Hajaj**, Itai Bendavid, Anat Goldstein, Eyal Robinson, and Pierre Singer. Using Machine-Learning to Assess the Prognostic Value of Early Enteral Feeding Intolerance in Critically Ill Patients: A Retrospective Study. *Nutrients*. 2023
17. Harel Berger, Amit Dvir, Enrico Mariconti, and **Chen Hajaj**. Breaking the structure of MaMaDroid. *Expert Systems with Applications*. 2023
18. Ofir Joseph, Avshalom Elmalech, and **Chen Hajaj**. Detecting Parallel Covert Data Transmission Channels in Video. *Electronics*. 2023

19. Or Haim Anidjar, Yannick Estève, **Chen Hajaj**, Amit Dvir, Itshak Lapidot. Speech and Multilingual Natural Language Framework for Speaker Change Detection and Diarization. *Expert Systems With Applications*. 2023
20. Adi Lichy, Ofek Bader, Ran Dubin, Amit Dvir, **Chen Hajaj**. When a RF Beats a CNN and GRU, Together - A Comparison of Deep Learning and Classical Machine Learning Approaches for Encrypted Malware Traffic Classification. *Computers & Security*. 2023
21. Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich, Corine Berger. The Online temporary learning groups in higher education: Interactions, compensation, and maximization of achievements in an Israeli case study. *Journal of Education Culture and Society*. 2023
22. Rotem Bar and **Chen Hajaj**. SimCSE for Encrypted Traffic Detection and Zero-Day Attack Detection. *IEEE ACCESS*. 2022
23. **Chen Hajaj**, Nitay Hason, and Amit Dvir. Less is More: Robust and Novel Features for Malicious Domain Detection. *Electronics*. 2022
24. Anat Goldstein and **Chen Hajaj**. The Hidden Conversion Funnel of Mobile vs. Desktop Consumers. *Electronic Commerce Research and Applications (ECRA)*. 2022
25. Corine Berger, Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich. Patterns of Activity in Temporary Digital Learning Groups, and Its Correlation With Academic Achievement Among University Students in an Israeli University. *Social Issues in Israel*. 2022
26. Harel Berger, **Chen Hajaj**, and Amit Dvir. Crystal Ball: From Innovative Attacks to Attack Effectiveness Classifier. *IEEE ACCESS*. 2021
27. Ron Hirschprung and **Chen Hajaj**. Prediction Model for the Spread of the COVID-19 Outbreak in the Global Environment. *Heliyon*. 2021
28. Or Anidjar, Itshak Lapidot, **Chen Hajaj**, Amit Dvir and Issachar Gilad. Hybrid Speech and Text Analysis Methods for Speaker Change Detection. *IEEE/ACM Transactions on Audio Speech and Language Processing*. 2021
29. **Chen Hajaj**, Zlatko Joveski, Sixie Yu, Yevgeniy Vorobeychik. Robust Coordination in Adversarial Social Networks: From Human Behavior to Agent-Based Modeling. *Network Science*. 2021
30. Amit Dvir, Angelos K. Marnerides, Ran Dubin, Nehor Golan, and **Chen Hajaj**. Encrypted Video Traffic Clustering Demystified. *Computers & Security*, 96:101854, 2020
31. **Chen Hajaj** and David Sarne. Selective Opportunity Disclosure at the Service of Strategic Information Platforms. *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(5):1133–1164, 2017
32. **Chen Hajaj**, Noam Hazon, and David Sarne. Enhancing Comparison Shopping Agents through Ordering and Gradual Information Disclosure. *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(3):696–714, 2017
33. **Chen Hajaj**, Noam Hazon, and David Sarne. Improving Comparison Shopping Agents' Competence through Selective Price Disclosure. *Electronic Commerce Research, and Applications (ECRA)*, 14(6):563–581, 2015

## REFEREED CONFERENCE PUBLICATIONS

1. **Chen Hajaj**, Wiam Meshleb, and Amit Dvir. When Evaluation Protocol Changes the Winner: Traffic Classification Beyond Architecture. In *IEEE Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, January 2027. To appear.
2. Liav Cohen, Amit Dvir, **Chen Hajaj**. FALCON: Few-shot ANN for Low-latency Classification in O-RAN Networks. In *IEEE International Workshop on Computer-Aided Modeling and Design of Communication Links and Networks (CAMAD)*. 2026
3. Hadas Sopher, **Chen Hajaj**, John S. Gero. Learning through concept generation: The case of academic and practitioner teaching in the design engineering studio. In *Design Computing and Cognition*. 2026

4. Eylon Yehiel, Anat Goldstein, **Chen Hajaj**. Product Pricing by Multidimensional Similarity: A Multimodal Attention Framework Using Image, Text, and Temporal Representations. In *the 20th ILAIS Conference*, Peres Academic Center, Israel, July 2026. **Best Paper Award (research-in-progress), second place.**
5. Udi Aharon, Amit Dvir, Ran Dubin, Revital Marbel, **Chen Hajaj**. Uncovering Microservice Faults: A Temporal Graph Approach to Root Cause Analysis. In *IEEE International Conference on Communications (ICC 2026)*, Glasgow, Scotland, UK, May 2026.
6. Meir Litman and **Chen Hajaj**. Cleaner Adversarial CAPTCHAs: Intelligent Targets and Precise Noise for Usable Security. In *The 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2026
7. Yehonatan Zion, Eyal Paz, Ran Dubin, Amit Dvir, **Chen Hajaj**. Quality of Experience Prediction for First Person Shooter Online Gaming: The Case Study of Call of Duty. In *IEEE Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, January 2026.
8. Revital Marbel, Yanir Victor Cohen, Ran Dubin, Amit Dvir, **Chen Hajaj**. Cloudy with a Chance of Anomalies: Dynamic Graph Neural Network for Early Detection of Cloud Services' User Anomalies. In *IEEE Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, January 2026.
9. Itay Meiri, Ran Dubin, Amit Dvir, **Chen Hajaj**. Out-Of-Distribution Is Not Magic: The Clash Between Rejection Rate and Model Success. In *the 20th Conference on Computer Science and Intelligence Systems (FedCSIS)*. 2025
10. Hadas Sopher, **Chen Hajaj**, John S. Gero. Sentiment Analysis of Student-Tutor Interactions in VR Design Crits. In *the 30th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, Tokyo, Japan. 2025
11. Simona Lisker, Ayelet Botman, **Chen Hajaj**, Ran Dubin, Amit Dvir. Optimized File Type Detection and One-Shot Retrieval. In *IEEE International Conference on Communications (ICC)*. 2025
12. Zohar Simhon, Matan Weiss, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. A New D-MAGIC: Dynamic Model for Cybersecurity Attack Detection Using GNNs into Clustering. In *IEEE International Conference on Communications (ICC)*. 2025
13. Yehonatan Zion, Porat Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. Enhancing Encrypted Internet Traffic Classification Through Advanced Data Augmentation Techniques. In *IEEE International Conference on Communications (ICC)*. 2025
14. Angelos Marnerides, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. PQClass: Classification of Post-Quantum Encryption Applications in Internet Traffic. In *IEEE International Conference on Communications (ICC)*. 2025
15. Hadas Sopher, **Chen Hajaj**, John S. Gero. Exploring the Role of Sentiment in Tutor-Student Interaction. The Case Study of CS and Architecture Formative Studio Critiques. In *IEEE Frontiers in Education Conference (FIE)*. 2024
16. Anat Goldstein and **Chen Hajaj**. Warm Recommendation: Enhancing Cold Start Recommendations Using Multimodal Product Representations. In *International conference of the Association for Information Systems (ICIS)*. 2024
17. Nathan Dillbary, Roi Yozevitch, Amit Dvir, Ran Dubin, and **Chen Hajaj**. Hidden in Time, Revealed in Frequency: Spectral Features and Multiresolution Analysis for Encrypted Internet Traffic Classification. In *IEEE Consumer Communications & Networking Conference (CCNC)*. 2024
18. Anat Goldstein and **Chen Hajaj**. Warming Up the Cold Start: A Multimodal Approach. In *Special Interest Group on Decision Support and Analytics (SIGDSA) Symposium*. 2023
19. Orit Raphaeli, Liran Statlander, **Chen Hajaj**, Itai Ben David, and Pierre Singer. Protein Intake and Clinical Outcomes of Critically Ill Patients. In *the 44th European Congress for Clinical Nutrition and Metabolism (ESPEN)*. 2022
20. Anat Goldstein and **Chen Hajaj**. Measuring Product Similarity: A Multidimensional Approach. In *the Eighteenth Symposium on Statistical Challenges in Electronic Commerce Research*. 2022

21. Ofek Bader, Adi Lichy, **Chen Hajaj**, Ran Dubin, and Amit Dvir. MalDIST: From Encrypted Traffic Classification to Malware Traffic Detection and Classification. In *IEEE Consumer Communications & Networking Conference (CCNC)*. 2022
22. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer, and Uri Gabbay. Using the Cardio-Vascular Index (CVRI) To Predict Mortality in Septic Shock. In *the 40th International Symposium on Intensive Care & Emergency Medicine*. 2022
23. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Support Early Prediction of Feeding Intolerance in Critically Ill Patients. In *Intensive Care Medicine Experimental*. 2021
24. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Compare Gastric Residual Volume Thresholds as Predictors of Clinical Outcomes in Critically Ill Patients. In *International Symposium of Intensive Care and Emergency Medicine*. 2021
25. Or Anidjar, Itshak Lapidot, **Chen Hajaj** and Amit Dvir. A Thousand Words are Worth More Than One Recording: Word-Embedding Based Speaker Change Detection. In *Interspeech*. 2021
26. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Feeding Intolerance as a Predictor of Clinical Outcomes in Critically Ill Patients: A Machine Learning Approach. In *the 43rd European Society for Clinical Nutrition and Metabolism (ESPEN)*. 2021
27. Uzi Ben Shalom, **Chen Hajaj**, Nitza Davidovich and Corinne Berger. Online temporary learning groups in higher education – Interactions, compensation, and maximization of achievements in an Israeli case-study. In *the 1st international e-conference on teacher education generation next: perspectives, opportunities & challenges*. 2021
28. Vitali Demianiuk, **Chen Hajaj**, and Kiril Kogan. PCL: Packet Classification with Limited Knowledge. *INFOCOM*. 2021
29. Anat Goldstein and **Chen Hajaj**. The Different Path to Purchase of Mobile and Desktop Consumers: Analyzing Consumers' Progress in the Conversion Funnel Using Hidden Markov Models. In *Statistical Conference in E-Commerce Research (SCECR)*. 2020
30. Nitay Hason, Amit Dvir, and **Chen Hajaj**. Robust Malicious URL Detection. In the *4th International Symposium on Cyber Security Cryptology and Machine Learning* (CSCML). 2020
31. Liang Tong, Bo Li, **Chen Hajaj**, Chaowei Xiao, Ning Zhang, Yevgeniy Vorobeychik. Improving Robustness of ML Classifiers against Realizable, Evasion Attacks Using Conserved Feature. In the *28th USENIX Security Symposium*. 2019
32. Yevgeniy Vorobeychik, Myrna Wooders, Greg Leo, Matthew Chambers, **Chen Hajaj**, Jian Lou, and Martin Van der Linden. Non-Cooperative Team Formation and a Team Formation Mechanism. In the *11th annual conference of the Israeli chapter of the game theory society*. 2019
33. **Chen Hajaj**, Sixie Yu, Zlatko Joveski, Yifan Guo, and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *the International Conference on Autonomous Agents and Multiagent Systems* (AAMAS). 2019
34. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin, and Daniel Fabbri. Crowdsourcing Clinical Chart Reviews. In *AMIA Joint Summits on Translational Science*. 2018
35. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial task assignment. In *International Joint Conference on Artificial Intelligence* and *European Conference on Artificial Intelligence* (IJCAI-ECAI). 2018
36. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin and Daniel Fabbri. A Crowdsourcing Framework for Medical Data Sets. In *American Medical Informatics Association* (AMIA). 2018
37. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Enhancing Crowdworkers' Vigilance. In *International Joint Conference on Artificial Intelligence* (IJCAI). 2017

38. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Extending Workers' Attention Span Through Dummy Sub-Goals. In *AAAI Conference on Human Computation and Crowdsourcing* (HCOMP). 2016. **Best paper finalist.**
39. **Chen Hajaj**, John P. Dickerson, Avinatan Hassidim, Tuomas Sandholm and David Sarne. Strategy-Proof and Efficient Kidney Exchange Using a Credit Mechanism. In *AAAI Conference on Artificial Intelligence* (AAAI). 2015
40. Tal Darom, Yaniv Gur, **Chen Hajaj** and Yosi Keller. Automatic Anatomical Shape Correspondence and Alignment Using Mesh Features. In *International Symposium on Biomedical Imaging* (ISBI). 2015
41. **Chen Hajaj** and David Sarne. Strategic Information Platforms - Selective Disclosure and The Price of "Free". In *ACM Conference on Economics and Computation* (EC). 2014
42. **Chen Hajaj**, Noam Hazon, and David Sarne. Ordering Effects and Belief Adjustment in the Use of Comparison Shopping Agents. In *AAAI Conference on Artificial Intelligence* (AAAI). 2014
43. **Chen Hajaj**, Noam Hazon, David Sarne, and Avshalom Elmalech. Search More, Disclose Less. In *AAAI Conference on Artificial Intelligence* (AAAI). 2013

**SHORT PAPERS AND REFEREED WORKSHOPS**

1. Brigitta R Schvarcz and **Chen Hajaj**. Exploring Student Perspectives of the Impact of VR in ESP. In *TESOL*. 2024
2. Bar Vinder, **Chen Hajaj**, and Idit Avrahami. A method for evaluating the performance of a novel percutaneous Left Ventricular Assist Device and optimizing its design. In *VPH 2022 PHYSIOLOGICAL HUMAN CONFERENCE*. 2022
3. Harel Berger, **Chen Hajaj**, and Amit Dvir. Evasion is not enough: A case study of Android malware. In the *4th International Symposium on Cyber Security Cryptology and Machine Learning* (CSCML). 2020
4. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial Randomized Task Assignment, 6th Day on Computational Game Theory. 2019
5. **Chen Hajaj**, Sixie Yu, Zlatko Joveski ,and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence*($AI^3$). 2018
6. Jian Lou, **Chen Hajaj**, Myrna Wooders and Yevgeniy Vorobeychik. Automated Mechanism Design for the Roommates Problem. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence* ($AI^3$). 2018
7. Jian Lou, Martin Van der Linden, Pranav Batra, **Chen Hajaj**, Greg Leo, Yevgeniy Vorobeychik and Myrna Wooders. Rotating Proposer Mechanisms for Team Formation. In *Workshop on Cooperative Games in Multiagent Systems* (CoopMAS).2017. **Most visionary paper.**
8. **Chen Hajaj**, David Sarne, and Lea Perets. Advanced Service Schemes for a Self-Interested Information Platform. In *International Conference on Autonomous Agents and Multiagent Systems* (AAMAS). 2014

**BOOK CHAPTERS**

Sofia Ceppi, Esther David, **Chen Hajaj**, Valentin Robu, and Ioannis A. Vetsikas, editors, Agent-Mediated Electronic Commerce: Designing Trading Strategies and Mechanisms for Electronic Markets, Springer, LNBIP 271, 2017.